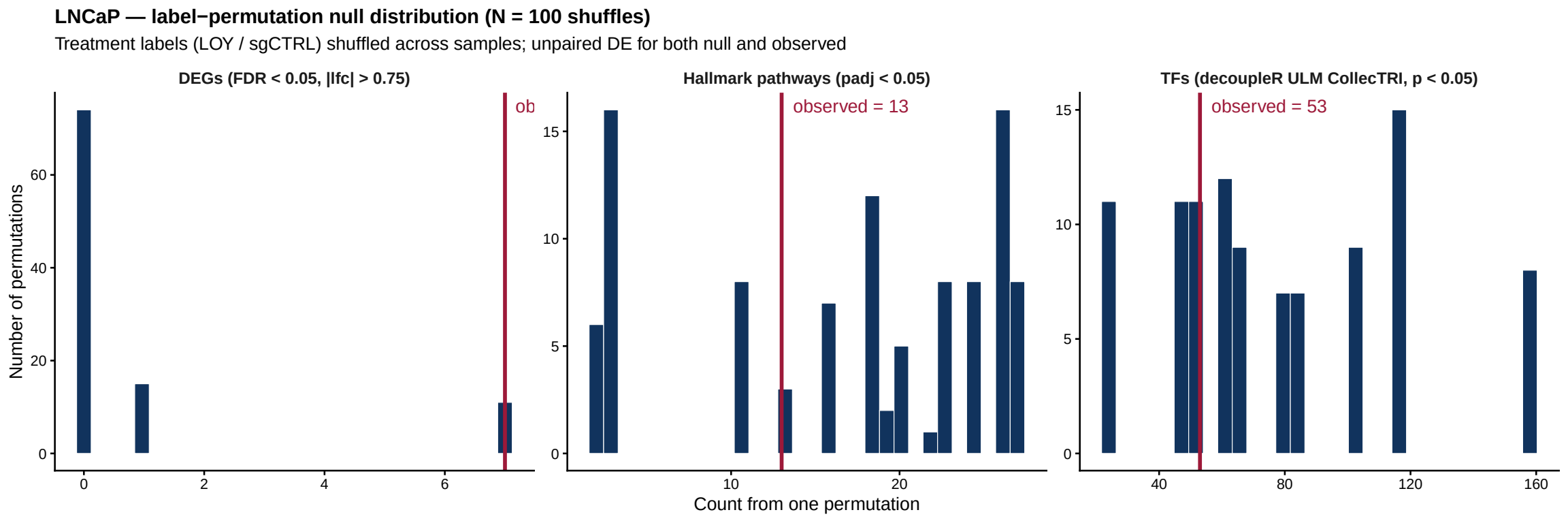
LNCaP — label−permutation null distribution (N = 100 shuffles)
Treatment labels (LOY / sgCTRL) shuffled across samples; unpaired DE for both null and observed
DEGs (FDR < 0.05, |lfc| > 0.75)
Hallmark pathways (padj < 0.05)
TFs (decoupleR ULM CollecTRI, p < 0.05)
ob
observed = 13
observed = 53
Number of permutations
Count from one permutation
0
20
40
60
0
2
4
6
0
5
10
15
10
20
0
5
10
15
40
80
120
160